Unit 7, Lesson 5: Factoring Trinomials

Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.

Learning Target

- [ ] I can factor trinomials.

7.5.1 Warm-Up: Walking Florida

1. Florida has the longest coastline, 1197 miles, in the contiguous United States. The distance from Key West to the farthest west Florida border on I-10 is 839 miles. The average person walks 3 miles per hour (mph).

   a. How many hours would it take someone walking 3 mph to walk from Key West to Florida's western border?

   b. How much longer, in days, would it take someone to walk along Florida's coastline than to walk from Key West to Florida's western border?

7.5.2 Exploration: Relating Terms and Factors

1. The steps for using the distributive property to multiply $(2x+3)(2x-2)$ are shown.

Step	$(2x+3)(2x-2)$	Result
Distribute $2x$ to $2x$	$(2x+3)(2x-2)$	$4x^2$
Distribute $2x$ to -2	$(2x+3)(2x-2)$	$-4x$
Distribute 3 to $2x$	$(2x+3)(2x-2)$	$6x$
Distribute 3 to -2	$(2x+3)(2x-2)$	-6

a. Based on the results in the table, what quadratic expression is the product of $(2x+3)(2x-2)$?

b. Tyra and Min-Joon were discussing how to find the x-term of the quadratic expression for $(2x+3)(2x-2)$.

Tyra	Min-Joon
The x-term of $(2x+3)(2x-2)$ can be quickly found by adding 3 and -2.	Looking at the table, the x-term of the quadratic expression is $2x$. I think you could just use what repeats.

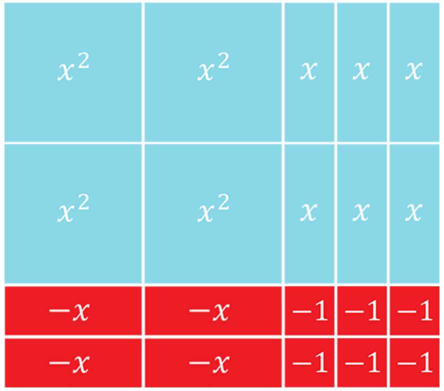
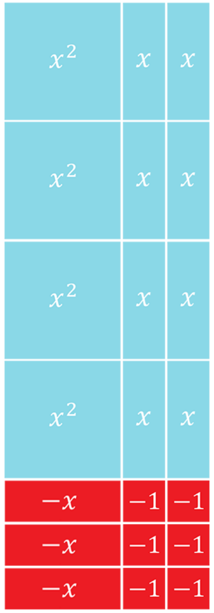
Discuss with your partner both students' strategies.

c. Both students have made a mistake. Describe to Tyra and Min-Joon how to find the x-term of the quadratic expression for $(2x+3)(2x-2)$.

d. Discuss with your partner whether or not the x-term of the quadratic expression $(4x-3)(x-2)$ is different from the x-term of the quadratic expression $(2x-3)(2x-2)$. Summarize your discussion.

2. Consider the trinomials $4x^2 + 2x - 6$ and $4x^2 + 5x - 6$.

   a. Use the algebra tiles to factor each trinomial.

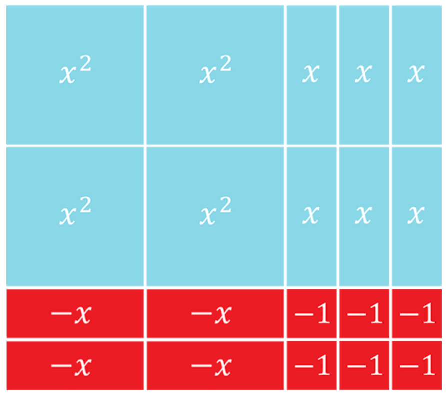
Trinomial	$4x^2 + 2x - 6$	$4x^2 + 5x - 6$
Algebra Tiles		
Factored Form		

   b. The trinomials have the same leading term, $4x^2$. List the two ways $4x^2$ is factored.

   c. List the factor pairs of -24.

   d. Which pair of factors of -24 add to 2?

   e. Discuss with your partner how the pair of factors of -24 that add to 2 is related to the factored form of $4x^2 + 2x - 6$. Show the factors along the algebra tiles.



7.5.3 Activity: Card Sort

1. Work with your partner to sort the cards. Write the letter of the matching card for each representation.

Quadratic Expression	Algebra Tiles	Area Model	Expanded Form	Factored Form
$6x^2 - x - 2$				
$6x^2 + x - 2$				
$6x^2 - 7x + 2$				
$6x^2 - 11x - 2$				
$6x^2 + 11x - 2$				
$6x^2 - 13x + 2$				

2. Describe one strategy you and your partner used to sort the cards.

7.5.4 Guided Instruction: Factoring Trinomials in the Form $ax^2 + bx + c$ when a has Multiple Factors

Factoring trinomials of the form $ax^2 + bx + c$, where the value of a has multiple factors, requires all of the factors of a to be considered when factoring using algebra tiles or the area model.

1. Complete the table for $10x^2 + x - 3$.

	____x	________
___x	$10x^2$	○ $1x$ ○ $-1x$ ○ $6x$ ○ $-6x$ ○ $2x$ ○ $-2x$ ○ $10x$ ○ $-10x$ ○ $3x$ ○ $-3x$ ○ $15x$ ○ $-15x$ ○ $5x$ ○ $-5x$ ○ $30x$ ○ $-30x$
____	○ $1x$ ○ $-1x$ ○ $6x$ ○ $-6x$ ○ $2x$ ○ $-2x$ ○ $10x$ ○ $-10x$ ○ $3x$ ○ $-3x$ ○ $15x$ ○ $-15x$ ○ $5x$ ○ $-5x$ ○ $30x$ ○ $-30x$	-3

2. Which expanded form of $10x^2 + x - 3$ could be used to factor the expression by grouping?

   ○ $10x^2 + 5x - 6x - 3$
   ○ $10x^2 - 5x + 6x - 3$
   ○ $10x^2 - 10x + 3x - 3$
   ○ $10x^2 + 10x - 3x - 3$

3. Use the expanded form to factor $10x^2 + x - 3$.

7.5.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the exploration and activity that I knew were …	Something I wondered during the exploration or activity was …	Something I learned during the exploration or activity was …

Unit 7, Lesson 6: Factoring by Grouping

Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.

Learning Target

☐ I can factor a polynomial by grouping.